

ISRO Hackathon: Forecasting Energetic Particle Radiation Environment

This guide provides a comprehensive roadmap for Problem Statement 14, focusing on data access, domain knowledge, technical methodology, and project scoping.

1. Data Access Points

Reliable data is the foundation of space weather forecasting. For this project, you will primarily use NASA's **Coordinated Data Analysis Web (CDAWeb)** for historical archives and ISRO's **PRADAN** portal for validation.

GOES >2 MeV Electron Flux (CDF Archives)

The GOES (Geostationary Operational Environmental Satellite) series provides the primary dataset for electron flux at GEO.

- **Source:** [NASA CDAWeb HTTP Archive](#)
- **Specific Dataset IDs:**
 - **GOES-16/17/18 (Recent):** GOES16_EPEAD_L2_MAGED (or similar Level 2 EPEAD data).
 - **Historical (GOES-13/15):** GOES13_EPS_1MIN , GOES15_EPS_1MIN .
- **Data Path Pattern:** [https://cdaweb.gsfc.nasa.gov/pub/data/goes/goes\[XX\]/eps/](https://cdaweb.gsfc.nasa.gov/pub/data/goes/goes[XX]/eps/) or [.../epeat/](https://cdaweb.gsfc.nasa.gov/pub/data/goes/goes[XX]/epeat/)

Wind Spacecraft Solar Wind Parameters

While the Wind spacecraft provides direct solar wind data, the **OMNI High-Resolution (HRO)** dataset is often preferred as it combines data from Wind, ACE, and DSCOVR, time-shifted to the Earth's bow shock.

- **Source:** [NASA CDAWeb OMNI Archive](#)
- **Dataset ID:** OMNI_HRO_1MIN (1-minute resolution) or OMNI_HRO_5MIN .
- **Parameters:** Solar wind speed (V_{sw}), IMF B_z (GSM coordinates), and Proton Density (N_p).
- **Direct Link:** [OMNI HRO 1-Min HTTP](#)

ISRO GRASP/GSAT Payload Data

The Geostationary Radiation Spectrometer (GRASP) on GSAT-19 provides crucial validation data at Indian longitudes.

- **Portal:** [ISRO PRADAN \(ISRO Science Data Archive\)](#)
 - **Access Status:** Publicly available but **requires registration**.
 - **Process:**
 1. Register for an account on the PRADAN portal.
 2. Search for "GSAT-19" or "GRASP" in the instrument/mission list.
 3. Submit a data request (usually approved automatically for research/educational purposes).
 4. Download the data in the provided format (often CDF or NetCDF).
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2. Core Domain Knowledge

How Earth's Radiation Belts Work

Earth is surrounded by two donut-shaped regions of trapped energetic particles called the **Van Allen Radiation Belts**. Geostationary satellites (at ~6.6 Earth radii) sit near the outer edge of the **Outer Belt**, which is primarily composed of high-energy electrons.

- **Flux Spikes:** These happen when the Sun "pushes" the radiation belt. Solar disturbances inject new particles and provide energy that accelerates existing "seed" electrons to relativistic speeds (>2 MeV).

Key Solar Wind Drivers

Not all solar wind variables are equal. Focus on these three:

1. **Solar Wind Speed (V_{sw}):** The most critical long-term driver. High-speed streams (from coronal holes) provide the energy for electron acceleration via Wave-Particle interactions.
2. **Southward IMF ($B_z < 0$):** When the Interplanetary Magnetic Field points south, it "reconnects" with Earth's magnetic field, opening a gate for solar wind energy to enter the magnetosphere.
3. **Solar Wind Density (N_p):** High density can actually "wipe out" the belts initially (magnetopause shadowing) before a spike occurs.

The 30-45 Minute Mechanism

The primary 30-45 minute lead time is driven by the **propagation delay** and **magnetospheric response**:

- Solar wind measured at the L1 point (Wind/ACE) takes ~30-60 minutes to reach Earth.

- Once it hits the magnetosphere, the energy transfer and initial particle injection happen relatively quickly, setting the stage for the flux changes seen at GEO.

3. Technical Approach

Spacepy vs. Cdflib

Feature	Spacepy (pyCDF)	Cdflib
Ease of Setup	Harder (Requires NASA C-Library installation)	Easier (Pure Python, <code>pip install cdflib</code>)
Performance	High (C-based)	Moderate
Interface	Very Pythonic (Dict-like)	Functional
Recommendation	Use cdflib for a 1-week hackathon to avoid environment setup headaches.	

Handling Spikes and Gaps

Space weather data is "noisy."

- **Missing Data:** Use **Linear Interpolation** for small gaps (<30 mins). For larger gaps, split the data into separate "sequences" rather than filling them with fake data.
- **Spikes/Outliers:** Apply a **Rolling Median Filter** to remove non-physical sensor noise while preserving real physical flux increases.
- **Normalization:** Log-transform the electron flux ($\log_{10}(Flux)$) before training, as it varies over several orders of magnitude.

Forecasting Approach: LSTM and Alternatives

- **LSTM (Long Short-Term Memory):** Excellent for this task because it "remembers" the history of solar wind inputs which have cumulative effects on flux.
- **Alternative (Transformer):** Better at capturing very long-range dependencies but requires significantly more data and tuning.
- **Recommendation:** Stick with **LSTM** or a **GRU** (Gated Recurrent Unit). They are easier to train on limited compute and highly effective for time-series regression with 11 years of data.

Evaluation: Heidke Skill Score (HSS)

HSS measures how much better your forecast is than a random guess or persistence (predicting that tomorrow will be exactly like today).

- **Range:** -1 to 1.
 - **1:** Perfect forecast.
 - **0:** No better than random/persistence.
 - **< 0:** Worse than random.
 - **"Good" Score:** In space weather, an HSS > 0.4 for multi-hour forecasts is considered very good. For 30-45 min, you should aim for **0.7 - 0.8**.
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4. Prior Work and Positioning

Existing Models

- **NOAA REFM (Relativistic Electron Forecast Model):** A simple linear filter model. It's the baseline to beat. It often struggles with the timing of rapid increases.
 - **SNB3GEO:** A neural-network-based model that uses solar wind and geomagnetic indices.
 - **USP (Unique Selling Proposition):** Most existing models use 1-hour averages. Your team can focus on **high-cadence prediction (1-min or 5-min resolution)** and the specific **validation at Indian longitudes** using the GRASP data, which is a rare focus in global models.
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5. Realistic Scope Assessment (1 Week)

Achievable Goals (Working Prototype)

- A pipeline that downloads 1 year of GOES and OMNI data, cleans it, and trains a simple LSTM.
- Evaluation of the 30-45 minute primary forecast.
- A basic visualization comparing "Predicted" vs "Observed" flux.

Planned Methodology (Presentation Only)

- **Multi-step forecasting (6hr, 12hr):** You likely won't have time to optimize these. Present the architecture for how you *would* do it (e.g., recursive forecasting or multi-

head output).

- **Full 11-year training:** Start with 1-2 years to prove the concept; mention the full 11-year scale-up as "Future Work."

Major Risks to Flag

1. **Data Access Delays:** PRADAN registration might take 24-48 hours. **Register NOW.**
 2. **CDF Parsing:** Spacepy installation often fails on Windows; use `cdflib` .
 3. **Compute Limits:** Training LSTMs on 11 years of 1-minute data will be slow on standard laptops. Use **Google Colab (GPU)** or downsample to 5-minute resolution.
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Sources

1. [NASA CDAWeb Documentation](#)
2. [NOAA SWPC GOES Electron Flux Products](#)
3. [ISRO PRADAN Portal](#)
4. [Heidke Skill Score Definition \(WMO\)](#)
5. [SpacePy vs. Cdflib Comparison](#)